

Self-Learning: NoSQL Data Models DBMS

> **Definition:** NoSQL (Not Only SQL) refers to non-relational database management systems designed for horizontal scalability, high-velocity big data, flexible schema-less data models, and high availability.

1. Detailed Technical Explanation

NoSQL databases trade traditional ACID strict transactional guarantees for high performance and horizontal scaling across distributed clusters (governed by the **CAP Theorem**).

CAP Theorem:

A distributed database system can guarantee at most **two out of three** properties simultaneously:

1. **C**

2. The Four Major NoSQL Data Models

| NoSQL Category | Data Model | Key Features | Popular Industry Tools | Use Cases |

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3. SQL vs NoSQL Architecture Comparison

| Feature | Relational SQL Databases | NoSQL Databases |

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4. Quick Recall Flow

NoSQL